

Global Healthcare Investment & Life Expectancy Dynamics (2000–2023)

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Tooling: Power BI, DAX, Python, Open Data (World Bank WDI)

1. Executive Summary

This project presents an end-to-end data analytics evaluation of the relationship between global healthcare expenditure, preventative care interventions, and long-term population health outcomes across 200+ countries from 2000 through 2023. Utilizing indicators from the World Bank Development Indicators (WDI) database, this study assesses key metrics:

- **Global Average Life Expectancy: 70.5 years**
- **Average Health Expenditure (% GDP): 6.2%**
- **Average Annual Under-5 Child Deaths: 355K** globally per aggregated period
- **Global Measles Immunization Coverage: 85.0%** (falling **10.0%** short of the **95% WHO Target**)

2. Analytical Insights & Core Findings

A. Preventative Care Drives Child Survival

- **Inverse Correlation:** Regional analysis demonstrates a strong inverse relationship between immunization coverage and child mortality.
- **Target Deficit:** High-income regions maintain immunization levels near or above the **95% WHO threshold**, resulting in minimal under-5 mortality. Conversely, Sub-Saharan Africa and South Asia lag behind immunization baselines, bearing the largest share of global child mortality.

B. Health Expenditure Efficiency & Longevity

- **Diminishing Returns on Spend:** Scatter plot analysis (*Health Spend % GDP vs Life Expectancy*) indicates that while higher-income nations allocate a greater percentage of GDP to health (often 8%–12%), spending alone does not guarantee superior longevity outcomes.
- **Outperformers:** Select lower-middle-income nations achieve strong life expectancies (>72 years) on modest expenditure budgets (<6% of GDP), highlighting structural delivery efficiency.

C. Global Resilience & Recovery Dynamics

- **Long-Term Trajectory:** Life expectancy exhibited steady, positive growth across all income tiers from 2000 through 2019.

- **Shock & Rebound:** The visual trend line captures a global decline around 2020 reflecting pandemic disruptions, followed by stabilization and recovery through 2023.

3. Strategic Recommendations

- i. **Prioritize Primary & Preventative Infrastructure:** Reaching universal 95% immunization coverage yields a significantly higher return on investment per life saved than late-stage clinical interventions.
- ii. **Target High-Mortality Regions:** Direct global healthcare aid and logistics toward closing the 10% global gap in herd immunity, particularly in Sub-Saharan Africa and South Asia.
- iii. **Benchmark Operational Efficiency:** Policy planners in lower-income tiers should model operational frameworks on middle-income nations achieving high longevity on constrained GDP allocations.